

# Emission and Price Optimization in Energy Production: A Data Science Approach

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**Abstract**—This project addresses the critical challenge of restructuring energy production portfolios to align with sustainability goals, focusing on combating climate change and ensuring energy security. The primary objective is to develop an optimization model that maximizes the share of renewable energy sources, minimizes greenhouse gas emissions, and minimizes electricity prices for end-users, while maintaining current energy production levels. This report details the problem formulation, the methodologies employed (Genetic Algorithms and Particle Swarm Optimization), a discussion of real-world applications and their relevance, an experimental evaluation of the model using data for Germany and France, and concluding remarks on the findings and potential impact. The model aims to provide decision-makers with viable energy policy alternatives under various scenarios.

## I. PROBLEM DESCRIPTION

The core of this project is an optimization problem aimed at guiding European countries in transitioning towards more sustainable energy policies. The goal is to restructure their energy production portfolios by simultaneously considering environmental and economic aspects.

The main objectives are:

- 1) **Minimizing Greenhouse Gas Emissions:** Reduce total greenhouse gas emissions by optimizing the mix of energy sources, utilizing average emission factors (e.g., kg CO<sub>2</sub>eq/MWh) for each source.
- 2) **Minimize Electricity Prices:** Prevent significant increases in electricity prices for end-users that might result from transitioning to different fuel types.
- 3) **Maintaining Total Energy Production:** Ensure that the total electricity production after optimization does not fall below the current levels for the country.
- 4) **Maximizing Renewable Energy Share:** Implicitly, by minimizing emissions and considering costs, the model aims to increase the proportion of renewable energy sources.

### A. Objective Function

The problem is formulated as a multi-objective optimization problem. The objective function to be minimized is:

$$\min_{x,y,z} f(x,y,z) = \alpha x + \beta z - \gamma y$$

where:

- $x$ : Annual average electricity price for household consumers.
- $y$ : Share of renewable energy in total energy production.
- $z$ : Greenhouse gas emissions from electricity generation.
- $\alpha, \beta, \gamma$ : Weighting coefficients reflecting the relative importance of each objective (price, emissions, renewable share).

Minimizing this function aims to decrease price ( $x$ ) and emissions ( $z$ ), while increasing the renewable energy share ( $y$ , due to the negative coefficient).

### B. Constraints

The optimization model adheres to the following constraints:

- 1) **Energy Constraint:** The total energy production should remain stable, without major drops or rises. Let  $T_{year}$  be the total energy production in a given year and  $T_{year+1}$  be the production in the subsequent year.  $\delta$  is the acceptable deviation.

$$|T_{year+1} - T_{year}| \leq \delta$$

Operationally, this is managed by ensuring the sum of new energy shares equals 100%, applied to the baseline total production  $T_{year}$ .

- 2) **Emission Constraint:** Greenhouse gas emissions should be reduced by at least 5 percent (in line with EU targets). Let  $z$  be the current greenhouse gas emissions and  $z_0$  be the emissions in the previous year.

$$z \leq 0.95 \cdot z_0$$

- 3) **Cost Constraint:** Electricity price should not increase by more than 10%. Let  $x$  be the current electricity price and  $x_0$  be the price in the previous year.

$$x \leq 1.10 \cdot x_0$$

### C. Solution Space

The solution space  $S \subseteq \mathbb{R}^3$  is defined as the set of all feasible values for the decision variables  $x$  (electricity price),

$y$  (renewable energy ratio), and  $z$  (greenhouse gas emission), satisfying the system:

$$S = \left\{ (x, y, z) \in \mathbb{R}^3 \left| \begin{array}{l} 0 \leq x \leq 1.10 \cdot x_0 \\ 0 \leq z \leq 0.95 \cdot z_0 \\ 0 \leq y \leq 1 \end{array} \right. \right\}$$

Additionally, the energy production constraint ( $|T_{year+1} - T_{year}| \leq \delta$ ) must be satisfied, which is handled by maintaining the total production level  $T_0$  (total energy production in the previous year) when applying the new optimized energy shares. The individual shares of energy sources are the primary decision variables in the algorithms, which then determine  $x$ ,  $y$ , and  $z$ .

## II. METHODOLOGY

The energy portfolio optimization problem is addressed using two distinct metaheuristic techniques: Genetic Algorithm (GA) and Particle Swarm Optimization (PSO). These algorithms were chosen for their efficacy in exploring complex, non-linear solution spaces characterized by multiple objectives and constraints. The primary goal is to identify an energy mix (represented as a vector of energy source shares) that minimizes a composite objective function, considering economic, environmental, and renewable energy targets, while adhering to defined operational and transitional constraints.

### A. Objective Function

The core of the optimization is an objective function,  $f(x_{\text{norm}}, y_{\text{norm}}, z_{\text{norm}})$ , which aims to be minimized. It is defined as:

$$f(x_{\text{norm}}, y_{\text{norm}}, z_{\text{norm}}) = \alpha \cdot x_{\text{norm}} + \beta \cdot z_{\text{norm}} - \gamma \cdot y_{\text{norm}}$$

where:

- $x_{\text{norm}}$  is the normalized average household electricity price, relative to the initial price  $x_0$ .
- $z_{\text{norm}}$  is the normalized total CO2 emissions from electricity generation, relative to the initial emissions  $z_0$ .
- $y_{\text{norm}}$  is the new share of renewable energy sources in the total energy mix (expressed as a ratio from 0 to 1).
- $\alpha, \beta, \gamma$  are weighting factors for price, emissions, and renewable share, respectively. In this study, their values are set as:  $\alpha = 2.0$ ,  $\beta = 0.8$ , and  $\gamma = 0.1$ .

The values  $x_0, y_0, z_0$ , and  $T_0$  (initial total production) are derived from country-specific data for the latest available year.

### B. Cost Modeling

The calculation of the new electricity price ( $x_{\text{new}}$ ) incorporates both operational and installation costs.

- **Operational Costs:** Unit operational costs for each energy source (in Euro per kWh) are defined for different countries (e.g., Germany - DE, France - FR). A default operational cost of 0.15 Euro/kWh is applied if a specific source cost is not available.
- **Installation Costs[9]:** Additional capacity incurs installation costs. These are calculated as an amortized capital expenditure (CAPEX) per kWh of increased production

for a given source. The amortized CAPEX is determined by the formula:

$$\text{Amortized CAPEX per kWh} = \frac{\text{CAPEX per kW}}{(\text{Capacity Factor} \cdot 8760 \frac{\text{hours}}{\text{year}} \cdot \text{Lifetime})}$$

Specific amortized costs are defined for various technologies (e.g., Solar PV, Wind, Nuclear). A default high installation cost of 30.0 Euro/kWh of increased share is used for sources not explicitly listed or if capacity factor or lifetime is non-positive. Installation costs are only applied to the positive difference between the new share and the initial share of an energy source.

The total system cost, comprising operational costs for all production and installation costs for new capacity, is divided by the total energy production ( $T_0$  in kWh) to derive  $x_{\text{new}}$ .

### C. Constraints and Penalties

The optimization is subject to several constraints. Violations of these constraints incur penalties, which are added to the objective function value.

- 1) **Price Constraint:** The new average electricity price ( $x_{\text{new}}$ ) should not exceed 1.12 times the initial price ( $x_0$ ).
- 2) **Emission Constraint:** The new total CO2 emissions ( $z_{\text{new}}$ ) should be at most 0.95 times the initial emissions ( $z_0$ ).
- 3) **Energy Share Change Constraints:** To ensure realistic transitions:
  - For sources with an initial share  $\geq 0.005$  (0.5%): The new share can deviate by at most  $\pm 30\%$  relative to its initial value (i.e.,  $s_{\text{initial}} \cdot (1 \pm 0.30)$ ) AND by at most  $\pm 0.015$  in absolute percentage points (i.e.,  $s_{\text{initial}} \pm 0.015$ ). The stricter of these two bounds (minimum of upper bounds, maximum of lower bounds) is applied.
  - For sources with an initial share  $< 0.005$  (0.5%): The new share must not exceed an absolute cap of 0.015 (1.5%).

Penalty calculations:

- Violations of the price and emission constraints are penalized using a weight of  $P_{\text{constraints}} = 35000$ .
- Violations of the normal share change constraints (for shares  $\geq 0.5\%$ ) are penalized using a weight of  $P_{\text{share\_normal}} = 5,000,000$ , with the exceeded amount raised to an exponent of 2.0.
- Violations of the cap for near-zero initial shares (shares  $< 0.5\%$ ) are penalized using a weight of  $P_{\text{share\_zero\_cap}} = P_{\text{share\_normal}} \cdot 10 = 50,000,000$ , with the exceeded amount raised to an exponent of 2.0.

The total fitness evaluated for each solution is  $\text{Fitness} = \text{objective\_value} + \text{penalty\_basic\_constraints} + \text{penalty\_share\_constraints}$ .

### D. Genetic Algorithm (GA) Implementation Details

The GA is configured with the following parameters and mechanisms:

- **Population Size:** 200 individuals.
- **Maximum Generations:** 300.
- **Initialization:** Individuals (energy portfolios) are created by generating random shares for each energy source, which are then normalized to sum to 1. All shares are  $\geq 0$ .
- **Fitness Evaluation:** As described in sections 2.1 and 2.3.
- **Selection:** Tournament selection is used. In each tournament, 7 individuals are randomly chosen from the population, and the one with the best (lowest) fitness value is selected as a parent.
- **Crossover:** Single-point crossover is applied with a probability of 0.85.
- **Elitism:** 10%.

#### E. Particle Swarm Optimization (PSO) Implementation Details

The PSO algorithm is configured as follows:

- **Number of Particles:** 100.
- **Maximum Iterations:** 300.
- **Initialization:** Particles are initialized with random positions.
- **Velocity and Position Update:** A particle  $i$ 's velocity  $v_i$  and position  $pos_i$  are updated at each iteration  $t$  using
  - Inertia weight  $w = 0.5$ .
  - Cognitive coefficient  $c_1 = 1.5$ .
  - Social coefficient  $c_2 = 1.5$ .
  - $r_1, r_2$  are random numbers uniformly distributed in  $[0, 1]$ , generated for each dimension.

The initial data for energy shares, prices, emissions, production levels, and specific cost parameters are primarily sourced from Eurostat, with supplementary data from relevant literature for certain cost figures.

### III. REAL-WORLD APPLICATION

Optimizing national energy portfolios is a critical task for governments and energy policymakers worldwide as they strive to meet climate targets, ensure energy security, and maintain economic competitiveness. Many international organizations and research groups develop models and scenarios to guide these transitions.

One prominent example is the work done by the International Renewable Energy Agency (IRENA). IRENA's REmap program, for instance, develops global renewable energy roadmaps to assess pathways for doubling the share of renewables in the global energy mix. [2, 9] These analyses often involve detailed modeling of energy systems, considering technological options, costs, and resource potentials at a broad scale. They provide high-level insights into the feasibility and impact of large-scale renewable energy deployment.

Another area of real-world application involves national or regional energy system models, often based on frameworks like TIMES (The Integrated MARKAL-EFOM System) or LEAP (Long-range Energy Alternatives Planning System). These models are used by governmental agencies and research institutions to explore future energy scenarios, assess policy

impacts, and develop long-term energy strategies. For example, the IEA-ETSAP (Energy Technology Systems Analysis Programme) facilitates the use and development of such models for energy policy analysis, considering aspects like energy security, environmental impact, and economic development. [7, 20] Such models can perform least-cost optimization of the energy system under various constraints, including emission reduction targets.

Our project shares the overarching goal of these real-world applications: to provide data-driven guidance for energy policy. However, it has distinct characteristics and contributions:

- **Granular Multi-Objective Optimization:** While large-scale models often focus on least-cost pathways, our project employs a multi-objective approach simultaneously targeting emission minimization, price minimization, and renewable share maximization, with specific, user-defined weights for each objective. This allows for a more nuanced exploration of trade-offs.
- **Focus on Metaheuristics for Policy Alternatives:** The use of Genetic Algorithms and Particle Swarm Optimization is particularly suited for exploring a wide range of policy alternatives rather than a single "optimal" path. These algorithms can identify multiple near-optimal solutions, offering flexibility to decision-makers.
- **Direct Application to Specific Country Contexts:** The model is designed to use up-to-date, country-specific data (initially for Germany and France using Eurostat data), allowing for tailored recommendations that reflect the current energy mix, economic conditions, and policy constraints of a particular nation.
- **Emphasis on Practical Constraints:** The inclusion of constraints such as limits on electricity price increases and specific emission reduction targets, alongside operational constraints on how quickly the share of an energy source can change, grounds the model in real-world policy considerations.

The importance of our project lies in its potential to serve as an adaptable and relatively agile tool for national decision-makers. Unlike massive global models, it can be more readily updated with new data and re-run to assess emerging scenarios or policy ideas, providing concrete, percentage-based adjustments to the energy portfolio. This can complement the insights from larger, more complex modeling efforts by offering a focused analysis on specific policy levers and their immediate trade-offs.

### IV. EXPERIMENTAL EVALUATION

The proposed optimization model was experimentally evaluated using data for Germany (DE) and France (FR) for the year 2022, which was the latest available year in the dataset used. Both Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) were applied to find optimal energy production portfolios.

### A. Algorithm Convergence

The convergence behavior of GA and PSO was tracked by plotting the best fitness value found by the algorithms over generations (for GA) or iterations (for PSO).

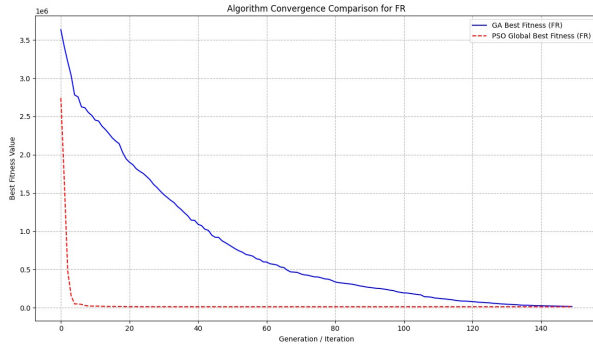


Fig. 1. Comparison of Genetic Algorithm and Particle Swarm Algorithm for France based on the best fitness values .

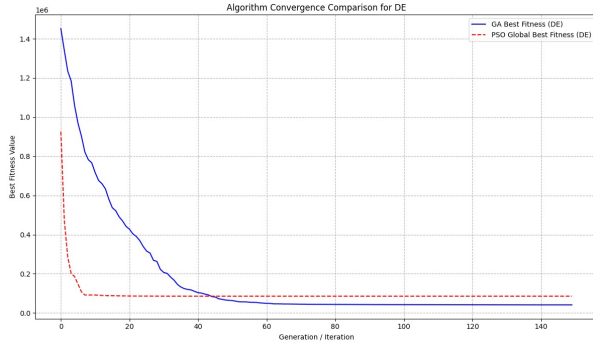


Fig. 2. Comparison of Genetic Algorithm and Particle Swarm Algorithm for Germany based on the best fitness values.

For both France (Figure 1) and Germany (Figure 2), PSO generally demonstrated faster convergence to a lower (better) fitness value compared to GA within the same number of iterations/generations. PSO quickly identified promising regions in the solution space, while GA showed a more gradual improvement.

### B. Optimized Energy Portfolios and Key Metrics

The following tables present the results obtained by GA and PSO for France and Germany. They show the initial energy shares, the optimized new shares, and the change in percentage points (p.p.) for key energy sources. Summary tables also provide an overview of the initial state and the achieved values for electricity price, renewable share, and CO2 emissions, along with their targets.

1) *France - Genetic Algorithm Results:* Table I shows the detailed energy share changes, and Table II summarizes the key metrics for France using GA.

2) *France - Particle Swarm Optimization Results:* Table III shows the detailed energy share changes, and Table IV summarizes the key metrics for France using PSO.

TABLE I  
FRANCE - GENETIC ALGORITHM: ENERGY SHARE CHANGES (%)

| Source                                      | Initial (%) | New (%) | Change (p.p.) |
|---------------------------------------------|-------------|---------|---------------|
| Nuclear_heat                                | 43.82       | 47.11   | 3.29          |
| Natural_gas                                 | 6.80        | 6.94    | 0.14          |
| Hydro                                       | 6.77        | 10.07   | 3.30          |
| Wind                                        | 5.68        | 8.95    | 3.27          |
| Solar_photovoltaic                          | 2.92        | 5.55    | 2.63          |
| Bioenergy                                   | 1.55        | 3.83    | 2.28          |
| Oil_and_petroleum_products...(excl biofuel) | 0.94        | 0.80    | -0.14         |
| Pumped_hydro_power                          | 0.82        | 2.88    | 2.05          |
| Primary_solid_biofuels                      | 0.68        | 2.61    | 1.94          |
| Other_bituminous_coal                       | 0.62        | 0.63    | 0.02          |
| Solid_fossil_fuels                          | 0.62        | 0.57    | -0.05         |
| Fuel_oil                                    | 0.56        | 0.69    | 0.13          |
| Biogases                                    | 0.45        | 1.65    | 1.21          |
| Non_renewable_waste                         | 0.33        | 0.00    | -0.33         |
| Non_renewable_municipal_waste               | 0.31        | 0.19    | -0.12         |
| Renewable_municipal_waste                   | 0.31        | 1.68    | 1.36          |
| Gas_oil_and_diesel_oil...(excl biofuel)     | 0.31        | 0.27    | -0.04         |
| Blast_furnace_gas                           | 0.25        | 0.00    | -0.25         |
| Manufactured_gases                          | 0.25        | 0.26    | 0.00          |
| Other_fuels_nec                             | 0.08        | 0.00    | -0.08         |
| Tide_wave_ocean                             | 0.07        | 1.69    | 1.62          |
| Refinery_gas                                | 0.06        | 0.00    | -0.06         |
| Industrial_waste_(non_renewable)            | 0.02        | 0.00    | -0.02         |
| Geothermal                                  | 0.02        | 1.69    | 1.67          |
| Other_oil_products_nec                      | 0.02        | 0.00    | -0.02         |
| Other_liquid_biofuels                       | 0.00        | 1.66    | 1.66          |
| Other_recovered_gases                       | 0.00        | 0.28    | 0.28          |

TABLE II  
FRANCE - GENETIC ALGORITHM: SUMMARY OF RESULTS

| Property             | Initial Value | Optimized Value | Target Value      |
|----------------------|---------------|-----------------|-------------------|
| Price (€/kWh)        | 0.2148        | 0.2230          | $\leq 0.2406$     |
| Renewable Share (%)  | 35.27         | 39.39           | Maximize          |
| CO2 Emission (Tonne) | 25,539,696    | 24,262,672      | $\leq 24,262,711$ |
| Best Fitness Value   | 11911.693896  |                 |                   |

TABLE III  
FRANCE - PARTICLE SWARM OPTIMIZATION: ENERGY SHARE CHANGES (%)

| Source                                      | Initial (%) | New (%) | Change (p.p.) |
|---------------------------------------------|-------------|---------|---------------|
| Nuclear_heat                                | 43.82       | 48.53   | 4.71          |
| Natural_gas                                 | 6.80        | 0.00    | -6.80         |
| Hydro                                       | 6.77        | 11.48   | 4.71          |
| Wind                                        | 5.68        | 10.39   | 4.71          |
| Solar_photovoltaic                          | 2.92        | 7.01    | 4.09          |
| Bioenergy                                   | 1.55        | 5.21    | 3.67          |
| Oil_and_petroleum_products...(excl biofuel) | 0.94        | 1.21    | 0.26          |
| Pumped_hydro_power                          | 0.82        | 4.27    | 3.44          |
| Primary_solid_biofuels                      | 0.68        | 4.08    | 3.41          |
| Other_bituminous_coal                       | 0.62        | 0.00    | -0.62         |
| Solid_fossil_fuels                          | 0.62        | 0.00    | -0.62         |
| Fuel_oil                                    | 0.56        | 0.00    | -0.56         |
| Biogases                                    | 0.45        | 1.82    | 1.37          |
| Non_renewable_waste                         | 0.33        | 0.00    | -0.33         |
| Non_renewable_municipal_waste               | 0.31        | 0.00    | -0.31         |
| Renewable_municipal_waste                   | 0.31        | 1.82    | 1.51          |
| Gas_oil_and_diesel_oil...(excl biofuel)     | 0.31        | 0.00    | -0.31         |
| Blast_furnace_gas                           | 0.25        | 0.29    | 0.03          |
| Manufactured_gases                          | 0.25        | 0.00    | -0.25         |
| Other_fuels_nec                             | 0.08        | 0.00    | -0.08         |
| Tide_wave_ocean                             | 0.07        | 0.00    | -0.07         |
| Refinery_gas                                | 0.06        | 0.10    | 0.04          |
| Industrial_waste_(non_renewable)            | 0.02        | 0.00    | -0.02         |
| Geothermal                                  | 0.02        | 1.82    | 1.80          |
| Other_oil_products_nec                      | 0.02        | 0.00    | -0.02         |
| Other_liquid_biofuels                       | 0.00        | 1.82    | 1.82          |
| Brown_coal_briquettes                       | 0.00        | 0.16    | 0.16          |

TABLE IV  
FRANCE - PARTICLE SWARM OPTIMIZATION: SUMMARY OF RESULTS

| Property             | Initial Value | Optimized Value | Target Value        |
|----------------------|---------------|-----------------|---------------------|
| Price (€/kWh)        | 0.2148        | 0.2406          | $\leq 0.2406$       |
| Renewable Share (%)  | 35.27         | 45.45           | Maximize            |
| CO2 Emission (Tonne) | 25,539,696    | 3,994,039       | $\leq 24, 262, 711$ |
| Best Fitness Value   | 52379.768918  |                 |                     |

3) *Germany - Genetic Algorithm Results:* Table V shows the detailed energy share changes, and Table VI summarizes the key metrics for Germany using GA.

TABLE V  
GERMANY - GENETIC ALGORITHM: ENERGY SHARE CHANGES (%)

| Source                                      | Initial (%) | New (%) | Change (p.p.) |
|---------------------------------------------|-------------|---------|---------------|
| Solid_fossil_fuels                          | 13.18       | 11.79   | -1.39         |
| Wind                                        | 9.14        | 14.27   | 5.13          |
| Lignite                                     | 8.45        | 8.39    | -0.06         |
| Natural_gas                                 | 6.12        | 6.77    | 0.65          |
| Solar_photovoltaic                          | 4.47        | 9.49    | 5.02          |
| Other_bituminous_coal                       | 4.46        | 4.66    | 0.19          |
| Bioenergy                                   | 3.63        | 8.56    | 4.93          |
| Nuclear_heat                                | 2.54        | 7.11    | 4.56          |
| Biogases                                    | 2.36        | 6.75    | 4.40          |
| Hydro                                       | 1.29        | 5.50    | 4.21          |
| Primary_solid_biofuels                      | 0.78        | 4.88    | 4.10          |
| Manufactured_gases                          | 0.70        | 0.89    | 0.19          |
| Non_renewable_waste                         | 0.47        | 0.00    | -0.47         |
| Blast_furnace_gas                           | 0.46        | 0.61    | 0.15          |
| Pumped_hydro_power                          | 0.44        | 1.90    | 1.46          |
| Non_renewable_municipal_waste               | 0.41        | 0.35    | -0.07         |
| Renewable_municipal_waste                   | 0.41        | 1.89    | 1.48          |
| Oil_and_petroleum_products...(excl biofuel) | 0.38        | 0.00    | -0.38         |
| Coking_coal                                 | 0.16        | 0.00    | -0.16         |
| Coke_oven_gas                               | 0.16        | 0.00    | -0.16         |
| Other_oil_products_nec                      | 0.15        | 0.00    | -0.15         |
| Gas_oil_and_diesel_oil...(excl biofuel)     | 0.09        | 0.08    | -0.01         |
| Other_fuels_nec                             | 0.09        | 0.00    | -0.09         |
| Fuel_oil                                    | 0.08        | 0.14    | 0.05          |
| Other_recovered_gases                       | 0.08        | 0.00    | -0.08         |
| Brown_coal_briquettes                       | 0.07        | 0.00    | -0.07         |
| Industrial_waste_(non_renewable)            | 0.06        | 0.16    | 0.10          |
| Refinery_gas                                | 0.05        | 0.00    | -0.05         |
| Anthracite                                  | 0.04        | 0.06    | 0.02          |
| Geothermal                                  | 0.02        | 1.90    | 1.88          |
| Other_liquid_biofuels                       | 0.01        | 1.95    | 1.94          |
| Tide_wave_ocean                             | 0.00        | 1.93    | 1.93          |

TABLE VI  
GERMANY - GENETIC ALGORITHM: SUMMARY OF RESULTS

| Property             | Initial Value | Optimized Value | Target Value         |
|----------------------|---------------|-----------------|----------------------|
| Price (€/kWh)        | 0.3318        | 0.3390          | $\leq 0.3716$        |
| Renewable Share (%)  | 38.32         | 57.12           | Maximize             |
| CO2 Emission (Tonne) | 216,552,402   | 205,712,970     | $\leq 205, 724, 782$ |
| Best Fitness Value   | 41184.714115  |                 |                      |

4) *Germany - Particle Swarm Optimization Results:* Table VII shows the detailed energy share changes, and Table VIII summarizes the key metrics for Germany using PSO.

### C. Discussion of Results

For both countries, the algorithms were able to find solutions that generally increased the share of renewable energy sources

TABLE VII  
GERMANY - PARTICLE SWARM OPTIMIZATION: ENERGY SHARE CHANGES (%)

| Source                                      | Initial (%) | New (%) | Change (p.p.) |
|---------------------------------------------|-------------|---------|---------------|
| Solid_fossil_fuels                          | 13.18       | 13.97   | 0.80          |
| Wind                                        | 9.14        | 14.70   | 5.56          |
| Lignite                                     | 8.45        | 8.42    | -0.03         |
| Natural_gas                                 | 6.12        | 7.67    | 1.55          |
| Solar_photovoltaic                          | 4.47        | 9.78    | 5.31          |
| Other_bituminous_coal                       | 4.46        | 3.11    | -1.35         |
| Bioenergy                                   | 3.63        | 8.75    | 5.12          |
| Nuclear_heat                                | 2.54        | 7.25    | 4.71          |
| Biogases                                    | 2.36        | 7.13    | 4.77          |
| Hydro                                       | 1.29        | 5.71    | 4.42          |
| Primary_solid_biofuels                      | 0.78        | 5.18    | 4.39          |
| Manufactured_gases                          | 0.70        | 0.43    | -0.27         |
| Non_renewable_waste                         | 0.47        | 0.00    | -0.47         |
| Blast_furnace_gas                           | 0.46        | 0.00    | -0.46         |
| Pumped_hydro_power                          | 0.44        | 1.91    | 1.47          |
| Non_renewable_municipal_waste               | 0.41        | 0.00    | -0.41         |
| Renewable_municipal_waste                   | 0.41        | 1.90    | 1.49          |
| Oil_and_petroleum_products...(excl biofuel) | 0.38        | 0.00    | -0.38         |
| Coking_coal                                 | 0.16        | 0.00    | -0.16         |
| Coke_oven_gas                               | 0.16        | 0.18    | 0.02          |
| Other_oil_products_nec                      | 0.15        | 0.00    | -0.15         |
| Gas_oil_and_diesel_oil...(excl biofuel)     | 0.09        | 0.07    | -0.02         |
| Other_fuels_nec                             | 0.09        | 0.02    | -0.06         |
| Fuel_oil                                    | 0.08        | 0.00    | -0.08         |
| Other_recovered_gases                       | 0.08        | 0.00    | -0.08         |
| Brown_coal_briquettes                       | 0.07        | 0.00    | -0.07         |
| Industrial_waste_(non_renewable)            | 0.06        | 0.00    | -0.06         |
| Refinery_gas                                | 0.05        | 0.00    | -0.05         |
| Anthracite                                  | 0.04        | 0.00    | -0.04         |
| Geothermal                                  | 0.02        | 1.90    | 1.89          |
| Tide_wave_ocean                             | 0.00        | 1.91    | 1.91          |

TABLE VIII  
GERMANY - PARTICLE SWARM OPTIMIZATION: SUMMARY OF RESULTS

| Property             | Initial Value | Optimized Value | Target Value         |
|----------------------|---------------|-----------------|----------------------|
| Price (€/kWh)        | 0.3318        | 0.3617          | $\leq 0.3716$        |
| Renewable Share (%)  | 38.32         | 56.96           | Maximize             |
| CO2 Emission (Tonne) | 216,552,402   | 205,724,505     | $\leq 205, 724, 782$ |
| Best Fitness Value   | 85772.653445  |                 |                      |

and adjusted the overall energy mix while attempting to stay within the price and emission constraints.

- **France:** Both GA and PSO found solutions that increased the renewable share and met the CO2 emission reduction target. The electricity price remained within the 10% increase limit. Pso reduced the co2 emission very much, although no similar situation was found in other results, the error here may be that it is possible to switch to a new energy immediately, this part was ignored because it would deepen the project too much, also since the installation costs for France could not be found, those of Germany were used, this may also be a problem, as a result, more detailed research is needed for total costs.
- **Germany:** Both algorithms significantly increased the renewable share. GA met the CO2 emission target and price constraint. PSO also met the price constraint but resulted in CO2 emissions higher than the target. GA achieved a better fitness value for Germany.

The changes in individual energy source shares vary between algorithms and countries, reflecting the complex trade-offs

involved. For instance, both algorithms generally increased shares for solar, wind, and biomass-related sources. The "Share Change Penalties" indicate that the algorithms had to balance the primary objectives with the constraints on how drastically individual shares could change, which is a practical consideration for policy implementation. PSO's faster convergence suggests it might be more efficient for quickly finding good solutions, though GA sometimes found solutions with slightly better adherence to all constraints or better objective values after more exploration, as seen in the German case.

## V. CONCLUSION

This project successfully developed and applied a multi-objective optimization model using Genetic Algorithms and Particle Swarm Optimization to restructure energy production portfolios for Germany and France. The model aimed to minimize greenhouse gas emissions and electricity prices while maximizing the share of renewable energy and maintaining total energy production levels.

The experimental results demonstrate that both GA and PSO can identify viable alternative energy portfolios that achieve significant improvements in terms of increased renewable energy share and reduced CO<sub>2</sub> emissions, while respecting predefined constraints on electricity price increases. For France, GA provided better overall solution meeting all targets, while for Germany, GA also outperformed PSO in meeting emission targets and achieving a better fitness score. PSO generally showed faster convergence.

The developed model serves as a valuable tool for energy policymakers, offering insights into the complex trade-offs involved in transitioning to more sustainable energy systems. By adjusting weighting factors and constraints, decision-makers can explore various scenarios and identify robust policy pathways. Future work could involve expanding the model to include more countries, incorporating more detailed technological and economic parameters, and exploring other advanced multi-objective optimization techniques. The inclusion of dynamic factors, such as technological learning curves and evolving carbon pricing mechanisms, could further enhance the model's predictive power and relevance.

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